

Workflow, Algorithm, Structure Reservoir Simulator Development

1) PREPARATION

Data Collections

- Model Geometry/Dimension
- Model Discretization (dx,dy,dz)
- **Model Properties**
 - Rock Properties (Grid Properties)
Porosity, PermX, PermY, PermZ, [PVMult, TransMult, NullGrid]
 - Rock – Fluid Interactions
Relative Permeabilities-Capillary Pressure **Table**
 - Rock Compressibility & its reference pressure
 - Fluid Properties [PVT **Table**, fluid density and its reference pressure]

2) Connection List

- **Structure Grid:**

- Connection list can be assigned as the E-W, S-N, T-B Cells of any evaluated Cells.

- **Unstructured Grid:**

- The indexing with I, j & k is no longer applicable. The connection should be calculated rigorously by detecting adjacent cells based any side of cell which used by two cells together.

- **Implement data structure**
to support Connection List,
which contains all information of
connection properties.

Connection Type

```
{  
  nCon      * number of Connection  
  ConTo ()  * Connection to Cell#  
  dCon ()   * Distance of Cells  
  Acon ()   * Area of Interface between connected cells  
  
  Trans ()  * Transmissibility between connected cells  
  TransMult() * Transmissibility between connected Cells  
}
```

3) Residual Calculation

Reference Formula:

$$Rp = \Delta a_p^{n+1} \Delta \Phi_p^{n+1} \pm q_p^{n+1} - \frac{1}{\Delta t} \left(\left(\frac{V_{pore} S_p}{B_p} \right)^{n+1} - \left(\frac{V_{pore} S_p}{B_p} \right)^n \right)$$

With **any** initial guess (iteration, $k=0$), usually using pressure at the beginning of time step:

Compute Rp using $p_k, Sw_k, Sg_k \rightarrow So_k = 1 - Sw_k - Sg_k$

Before compute **Rp** , you need to compute the PVT properties and rock properties using as function of guess parameters (p_k, Sw_k, Sg_k).

4a) Jacobian of Residual function

Reference Equation:

$$R_p = \Delta a_p^{n+1} \Delta \Phi_p^{n+1} \pm q_p^{n+1} - \frac{1}{\Delta t} \left(\left(\frac{V_{pore} S_p}{B_p} \right)^{n+1} - \left(\frac{V_{pore} S_p}{B_p} \right)^n \right)$$

Ideally the Jacobian is calculated using analytical method. Since it is quite challenging, thus we employ numerical method using perturbation.

$$\frac{\partial R_p^k}{\partial p_k} = \frac{R_p(p_k + \Delta p, Sw_k, Sg_k, So_k) - R_p(p_k, Sw_k, Sg_k, So_k)}{\Delta p}$$

$$\frac{\partial R_p^k}{\partial Sw_k} = \frac{R_p(p_k, Sw_k + \Delta Sw, Sg_k, So_k^*) - R_p(p_k, Sw_k, Sg_k, So_k)}{\Delta Sw}$$

$$\frac{\partial R_p^k}{\partial Sg_k} = \frac{R_p(p_k, Sw_k, Sg_k + \Delta Sg, So_k^*) - R_p(p_k, Sw_k, Sg_k, So_k)}{\Delta Sg}$$

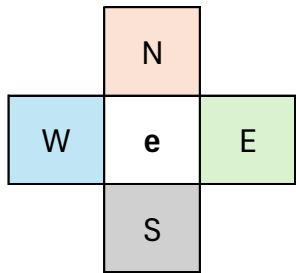
Remember:

Compute the fluid & rock properties at $p_k + dp$, $s + ds$ as well

4b) Jacobian of Residual function

Reference Equation:

$$R_p = \Delta a_p^{n+1} \Delta \Phi_p^{n+1} \pm q_p^{n+1} - \frac{1}{\Delta t} \left(\left(\frac{V_{pore} S_p}{B_p} \right)^{n+1} - \left(\frac{V_{pore} S_p}{B_p} \right)^n \right)$$



if we perturb a parameter of cell e (e.g the pressure of evaluated Cell), then beside the residual of Cell e, we should be able to compute the derivative of Residual of any connected cells with e (W,E,N,S,T,B) of all the phases.

Thus, if a cell is connected with 6 other cells and the problem is 3-phase (o,w,g), there will be $3 + 3*6 = 3*(1+nConnections)$ derivative of R_p with respect to each parameter. If we have 3 parameters (p, Sw, Sg), then the whole derivatives (total elements or non-zero element) as many as $3*3*(1+nConnection)$.

The general formula: $nPhases * nParams * (1 + nConnections)$.

Note: in Black oil model: $nPhases = nParams$

1	2	3
4	5	6

4c) Jacobian of Residual function

		1			2			3			4			5			6		
		p	Sw	Sg	p	Sw	Sg	p	Sw	Sg	p	Sw	Sg	p	Sw	Sg	p	Sw	Sg
1	o	dRo1dp1	dRo1dSw1	dRo1dSg1	dRo1dp2	dRo1dSw2	dRo1dSg2				dRo1dp4	dRo1dSw4	dRo1dSg4						
	w	dRw1dp1	dRw1dSw1	dRw1dSg1	dRw1dp2	dRw1dSw2	dRw1dSg2				dRw1dp4	dRw1dSw4	dRw1dSg4						
	g	dRg1dp1	dRg1dSw1	dRg1dSg1	dRg1dp2	dRg1dSw2	dRg1dSg2				dRg1dp4	dRg1dSw4	dRg1dSg4						
2	o	dRo2dp1	dRo2dSw1	dRo2dSg1	dRo2dp2	dRo2dSw2	dRo2dSg2	dRo2dp3	dRo2dSw3	dRo2dSg3				dRo2dp5	dRo2dSw5	dRo2dSg5			
	w	dRw2dp1	dRw2dSw1	dRw2dSg1	dRw2dp2	dRw2dSw2	dRw2dSg2	dRw2dp3	dRw2dSw3	dRw2dSg3				dRw2dp5	dRw2dSw5	dRw2dSg5			
	g	dRg2dp1	dRg2dSw1	dRg2dSg1	dRg2dp2	dRg2dSw2	dRg2dSg2	dRg2dp3	dRg2dSw3	dRg2dSg3				dRg2dp5	dRg2dSw5	dRg2dSg5			
3	o				dRo3dp2	dRo3dSw2	dRo3dSg2	dRo3dp3	dRo3dSw3	dRo3dSg3							dRo3dp6	dRo3dSw6	dRo3dSg6
	w				dRw3dp2	dRw3dSw2	dRw3dSg2	dRw3dp3	dRw3dSw3	dRw3dSg3							dRw3dp6	dRw3dSw6	dRw3dSg6
	g				dRg3dp2	dRg3dSw2	dRg3dSg2	dRg3dp3	dRg3dSw3	dRg3dSg3							dRg3dp6	dRg3dSw6	dRg3dSg6
4	o	dRo4dp1	dRo4dSw1	dRo4dSg1							dRo4dp4	dRo4dSw4	dRo4dSg4	dRo4dp5	dRo4dSw5	dRo4dSg5			
	w	dRw4dp1	dRw4dSw1	dRw4dSg1							dRw4dp4	dRw4dSw4	dRw4dSg4	dRw4dp5	dRw4dSw5	dRw4dSg5			
	g	dRg4dp1	dRg4dSw1	dRg4dSg1							dRg4dp4	dRg4dSw4	dRg4dSg4	dRg4dp5	dRg4dSw5	dRg4dSg5			
5	o				dRo5dp2	dRo5dSw2	dRo5dSg2				dRo5dp4	dRo5dSw4	dRo5dSg4	dRo5dp5	dRo5dSw5	dRo5dSg5	dRo5dp6	dRo5dSw6	dRo5dSg6
	w				dRw5dp2	dRw5dSw2	dRw5dSg2				dRw5dp4	dRw5dSw4	dRw5dSg4	dRw5dp5	dRw5dSw5	dRw5dSg5	dRw5dp6	dRw5dSw6	dRw5dSg6
	g				dRg5dp2	dRg5dSw2	dRg5dSg2				dRg5dp4	dRg5dSw4	dRg5dSg4	dRg5dp5	dRg5dSw5	dRg5dSg5	dRg5dp6	dRg5dSw6	dRg5dSg6
6	o							dRo6dp3	dRo6dSw3	dRo6dSg3				dRo6dp5	dRo6dSw5	dRo6dSg5	dRo6dp6	dRo6dSw6	dRo6dSg6
	w							dRw6dp3	dRw6dSw3	dRw6dSg3				dRw6dp5	dRw6dSw5	dRw6dSg5	dRw6dp6	dRw6dSw6	dRw6dSg6
	g							dRg6dp3	dRg6dSw3	dRg6dSg3				dRg6dp5	dRg6dSw5	dRg6dSg5	dRg6dp6	dRg6dSw6	dRg6dSg6

5) Update Iteration: $k \rightarrow k+1$

Update the value of p , S_w , S_g & S_o for iteration $k+1$ is obtained performing Newton Iteration Method.

$$J^k = \begin{bmatrix} \frac{\partial r_o}{\partial p} & \frac{\partial r_o}{\partial S_w} & \frac{\partial r_o}{\partial S_g} \\ \frac{\partial r_g}{\partial p} & \frac{\partial r_g}{\partial S_w} & \frac{\partial r_g}{\partial S_g} \\ \frac{\partial r_w}{\partial p} & \frac{\partial r_w}{\partial S_w} & \frac{\partial r_w}{\partial S_g} \end{bmatrix}^k$$

$$J^k [\Delta m]^{k+1} = -[r]^k$$

$$\Delta m^{k+1} = \begin{bmatrix} dp \\ dS_w \\ dS_g \end{bmatrix}^{k+1} = (J^k)^{-1} (-r^k)$$

$$\begin{bmatrix} p \\ S_w \\ S_g \end{bmatrix}^{k+1} = \begin{bmatrix} p \\ S_w \\ S_g \end{bmatrix}^k + \begin{bmatrix} dp \\ dS_w \\ dS_g \end{bmatrix}^{k+1}$$

6) Check Residual and Numerical Constraints

- Constraints:
 1. Number of Newton Iteration (MaxNewton)
 2. Maximum Residual Error (applied for each cell): MaxResidErr
 3. Max parameter Changes per step: (dpMax, dSwMax, dSgMax)

If all constraints are achieved, then $\text{time} = \text{time} + \text{delttime}$.

If not, $\text{delttime} = \lambda \text{ deltime}$ (with $\lambda < 1$ and $\lambda > 0$)